

TRUST & RELIABILITY LAYER FOR CLINICAL DECISION SUPPORT

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Background of Research

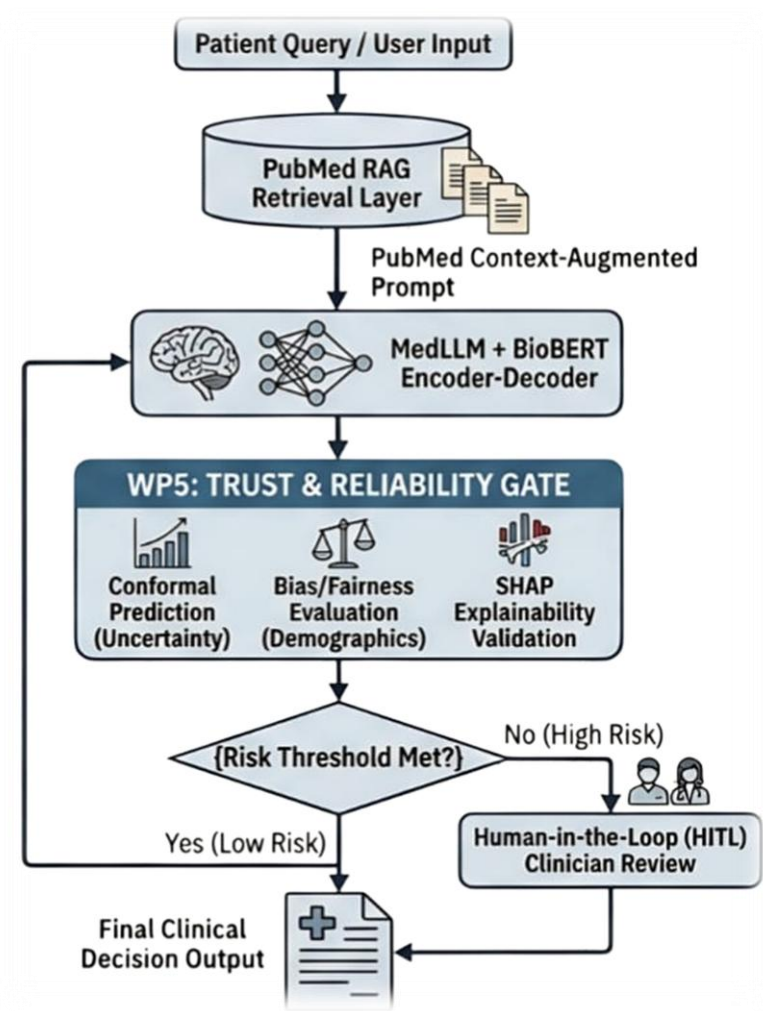
Healthcare LLMs support clinical reasoning but are limited by hallucinations, poor confidence calibration, and lack of transparency, reducing trust and patient safety [2]. Reliable uncertainty estimation and human oversight are essential for safe deployment [1,4].

Critical Review of Existing Works

- Existing CDSS tools lack reliable calibration and transparency [1,2].
- SoftMax confidence is poorly calibrated in clinical models [1]
 - LLMs can generate biased or unsafe medical outputs [5]
 - Models may be overconfident despite incorrect predictions[1,3]
 - Limited fairness evaluation leads to biased outcomes [5]
 - Human oversight is often absent in deployment systems [4]

Limitation	Clinical Impact
Hallucinations	Unsafe clinical outputs
Overconfident predictions	Misdiagnosis risk
Poor calibration (SoftMax)	Unreliable confidence
Bias in training data	Unequal healthcare outcomes

TrustMedLLM System Architecture



Proposed Trust & Reliability Framework

TrustMedLLM integrates conformal prediction for calibrated uncertainty estimation, fairness evaluation for demographic bias detection, and HITL clinician review for high-risk predictions. Low-risk outputs proceed directly to clinical decision output, while high-risk outputs are flagged for clinician review. OOD detection flags unfamiliar inputs to strengthen patient safety [1,4,5].

Key Novelties & Contributions

- CP + HITL for safe clinical decisions [1,4]
- Unified fairness–uncertainty framework [1,5]
- SHAP-driven explainable predictions
- Trust-aware reliable AI pipeline

References

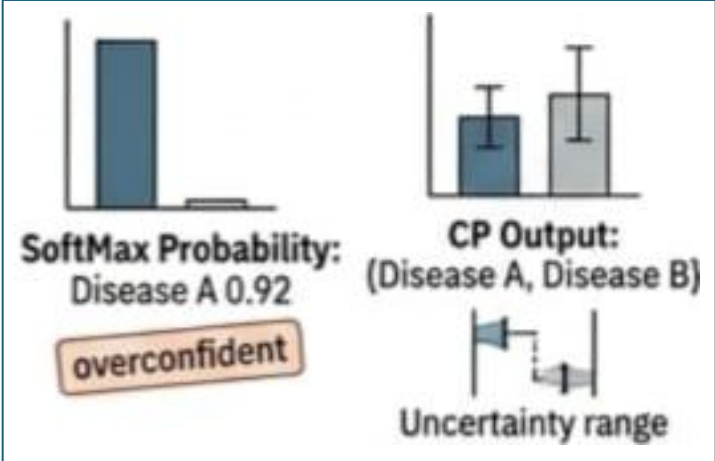
[1] Angelopoulos & Bates, 2022.
[2] Lewis et al., 2021.
[3] Cresswell et al., 2024.
[4] Wu et al., 2021.
[5] Mehrabi et al., 2021.

AI Use Declaration

AI tools used for figure generation only. All content and design decisions independently developed and verified by the author.

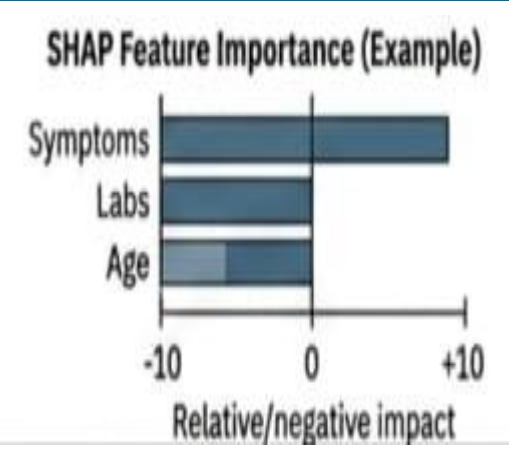
SoftMax vs Conformal Prediction

SoftMax is overconfident. CP provides valid confidence estimates [1,3].



Shap Explanation Block

SHAP identifies how clinical features influence predictions.



Expected Outcomes & Benefits

- Improved clinical decision safety through calibrated uncertainty estimation [1]
- Enhanced clinician trust via HITL review and explainable outputs [4]
- Reduced bias leading to fairer healthcare predictions [5]
- More reliable AI-assisted diagnostics with lower risk of unsafe ML outputs [2]
- Beneficiaries: clinicians, patients, and healthcare systems